# Mutual Fund Analytics Platform

## Final Project Report

### Executive Summary

#### Project Objective

The objective of this project is to design and implement a comprehensive Mutual Fund Analytics Platform that collects, processes, analyzes, and visualizes mutual fund and investor data. The system provides insights into fund performance, investor behavior, portfolio composition, and risk metrics through an interactive analytics dashboard.

#### Data Analyzed

The project utilizes multiple datasets representing different aspects of mutual fund operations:

- Fund Master Data
- NAV History
- Assets Under Management (AUM)
- SIP Inflows
- Investor Transactions
- Portfolio Holdings
- Benchmark Index Data

#### Key Outcomes

The project successfully:

- Built a complete ETL pipeline for data processing.
- Cleaned and standardized multiple datasets.
- Loaded processed data into SQLite.
- Performed exploratory data analysis (EDA).
- Calculated advanced performance metrics.
- Developed Power BI dashboards for visualization.
- Generated actionable insights for fund management and investor analysis.

---

# 2. Data Sources

The project uses the following datasets:

## Fund Master

Contains basic information about mutual funds including:

- Fund ID
- Fund Name
- Category
- Fund House
- Launch Date

## NAV History

Stores historical Net Asset Value records.

Fields:

- Fund ID
- Date
- NAV

Purpose:

- Trend Analysis
- Return Calculation
- Performance Measurement

## AUM Data

Contains Assets Under Management information.

Fields:

- Fund ID
- Date
- AUM Value

Purpose:

- Growth Analysis
- Fund Size Comparison

## SIP Inflows

Records monthly SIP investments.

Fields:

- Fund ID
- Month
- SIP Amount

Purpose:

- Investment Trend Analysis
- Retail Participation Analysis

## Investor Transactions

Stores investor purchase and redemption records.

Fields:

- Investor ID
- Gender
- Age
- Transaction Type
- Amount

Purpose:

- Investor Profiling
- Behavioral Analysis

## Portfolio Holdings

Contains security allocation details.

Fields:

- Security Name
- Sector
- Weight

Purpose:

- Portfolio Diversification Analysis
- Sector Exposure Analysis

## Benchmark Indices

Historical benchmark performance data used for:

- Alpha Calculation
- Beta Calculation
- Relative Performance Analysis

---

# 3. ETL Design

## Overview

The project follows an ETL architecture to transform raw data into analytical insights.

### ETL Workflow

Raw CSV Files ↓ Data Cleaning ↓ Processed CSV Files ↓ SQLite Database ↓ Analytics Layer ↓ Power BI Dashboard

### Extract Phase

Data is collected from multiple CSV files.

Tasks performed:

- Reading raw datasets
- Data validation
- Data type verification

### Transform Phase

Data cleaning operations include:

- Missing value handling
- Duplicate removal
- Date formatting
- Data standardization
- Outlier checks

### Load Phase

Processed datasets are loaded into SQLite.

Benefits:

- Centralized storage
- Faster querying
- Structured analytics

### Architecture Diagram

(Insert ETL Architecture Screenshot Here)

---

# 4. Exploratory Data Analysis (EDA)

## NAV Trend Analysis

(Insert NAV Trend Screenshot)

### Findings

- NAV values show consistent growth over time.
- Short-term volatility exists during market fluctuations.
- Long-term trend remains positive.

---

## Age Distribution Analysis

(Insert Age Distribution Screenshot)

### Findings

- Majority of investors fall between 25–45 years.
- Younger investors prefer SIP investments.
- Senior investors contribute larger investment amounts.

---

## Gender Distribution Analysis

(Insert Gender Distribution Screenshot)

### Findings

- Male investors represent a larger share.
- Female investor participation is increasing.
- Investment patterns differ across demographics.

---

## SIP Analysis

(Insert SIP Analysis Screenshot)

### Findings

- SIP inflows exhibit steady growth.
- Market corrections do not significantly affect SIP contributions.
- SIP remains the preferred long-term investment mode.

---

## T30/B30 Distribution

(Insert T30/B30 Distribution Screenshot)

### Findings

- T30 cities contribute the majority of investments.
- B30 participation shows growing adoption.
- Expansion opportunities exist in smaller cities.

---

# 5. Performance Analysis

## CAGR (Compound Annual Growth Rate)

Measures annualized investment growth.

### Interpretation

- Higher CAGR indicates stronger long-term performance.
- Useful for comparing mutual funds across categories.

---

## Sharpe Ratio

Measures risk-adjusted returns.

### Interpretation

- Higher Sharpe Ratio indicates better reward per unit of risk.
- Helps evaluate fund efficiency.

---

## Sortino Ratio

Measures downside-risk-adjusted returns.

### Interpretation

- Focuses only on negative volatility.
- Preferred for risk-sensitive investors.

---

## Maximum Drawdown

Measures largest decline from peak value.

### Interpretation

- Lower drawdown indicates better downside protection.
- Important during market downturns.

---

## Alpha and Beta

### Alpha

Measures excess return over benchmark.

Positive Alpha: Fund outperforms benchmark.

### Beta

Measures market sensitivity.

Beta > 1: Higher volatility than market.

Beta < 1: Lower volatility than market.

---

## VaR (Value at Risk)

Estimates potential portfolio loss under normal conditions.

### Interpretation

- Helps quantify investment risk.
- Widely used in portfolio management.

---

### CVaR (Conditional Value at Risk)

Measures expected loss beyond VaR threshold.

#### Interpretation

- Provides insight into extreme downside risk.
- Useful for stress testing.

---

# 6. Dashboard Screenshots

## Overview Dashboard

(Insert Overview Dashboard Screenshot)

Features:

- Total AUM
- Total Investors
- Total SIP Inflows
- Fund Distribution

---

## Fund Performance Dashboard

(Insert Fund Performance Screenshot)

Features:

- NAV Trends
- CAGR Analysis
- Risk Metrics
- Benchmark Comparison

---

## Investor Dashboard

(Insert Investor Dashboard Screenshot)

Features:

- Age Distribution

- Gender Analysis
- Transaction Analysis
- Geographic Distribution

---

# 7. Limitations

The project has the following limitations:

1. Simulated datasets were used for demonstration purposes.

2. Historical data covers a limited time period.

3. No live market feeds are integrated.

4. Risk metrics rely on available sample data.

5. Real-world market events are not fully represented.

---

# 8. Recommendations

Future enhancements include:

## Real-Time Data Integration

- Connect live NAV APIs.
- Update fund performance automatically.

## Predictive Analytics

- Forecast NAV movements.
- Predict investor behavior.

## Portfolio Optimization

- Recommend optimal asset allocation.
- Improve diversification strategies.

## Risk Forecasting

- Implement advanced risk models.
- Perform scenario analysis and stress testing.

### Cloud Deployment

- Deploy ETL pipeline in cloud environment.
- Enable automated scheduling and monitoring.

---

# Conclusion

The Mutual Fund Analytics Platform successfully demonstrates end-to-end data engineering, analytics, and visualization capabilities. By integrating ETL processes, performance analytics, risk assessment, and dashboard reporting, the project provides a strong foundation for data-driven mutual fund management and future enhancements in financial analytics.